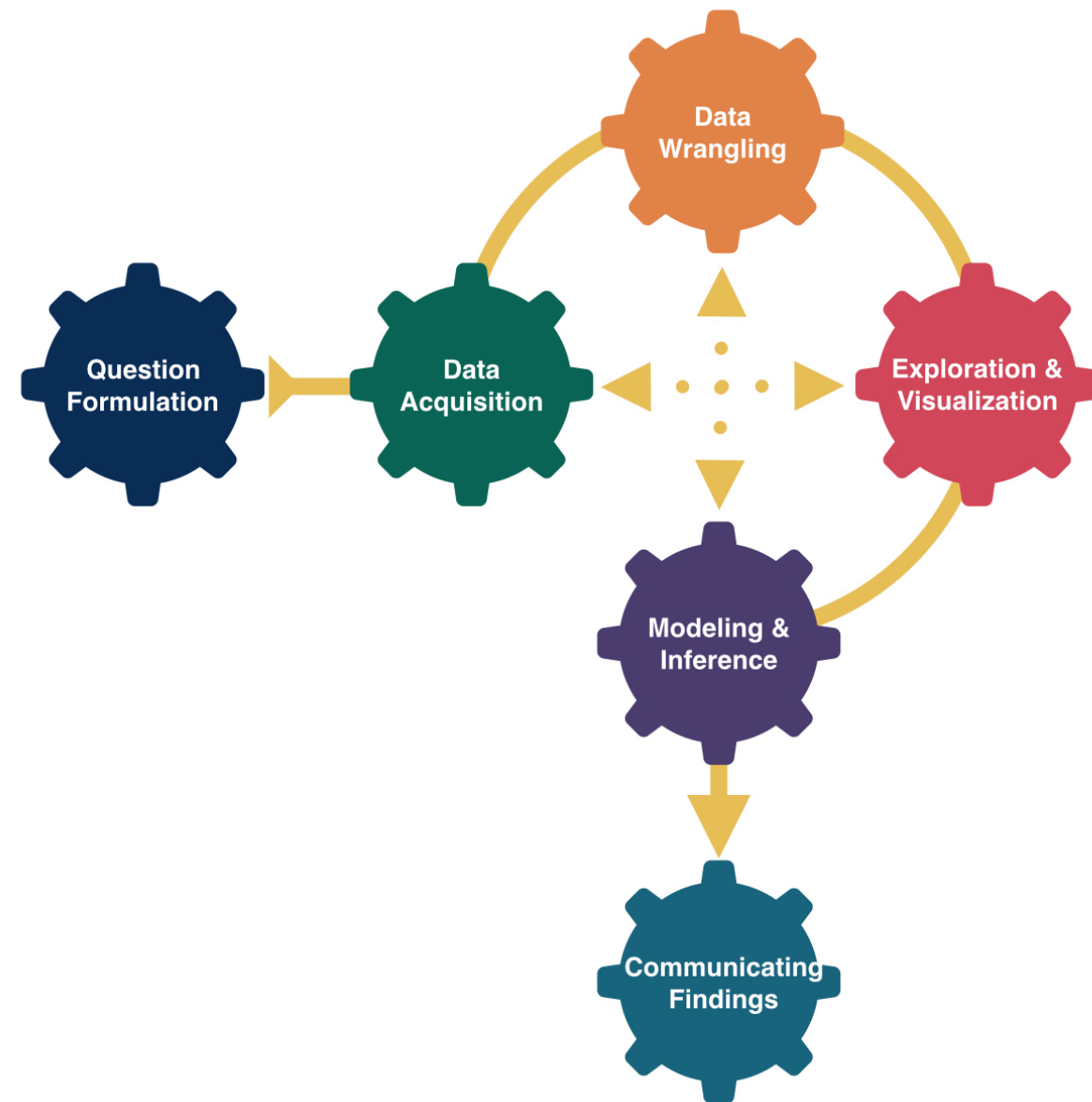




Logistic Regression + Classification Trees

Kelly McConville

DATA 100

Week 13 | Spring 2026

Announcements

- Don't forget to get your data science activity credit.

Goals for Today

- Logistic Regression
- Classification Trees
- Installing R and RStudio on your computer

Models for Categorical Response Variables

Logistic Regression

Response variable: A **categorical** variable with **two** categories

$$Y = \begin{cases} 1 & \text{success} \\ 0 & \text{failure} \end{cases}$$

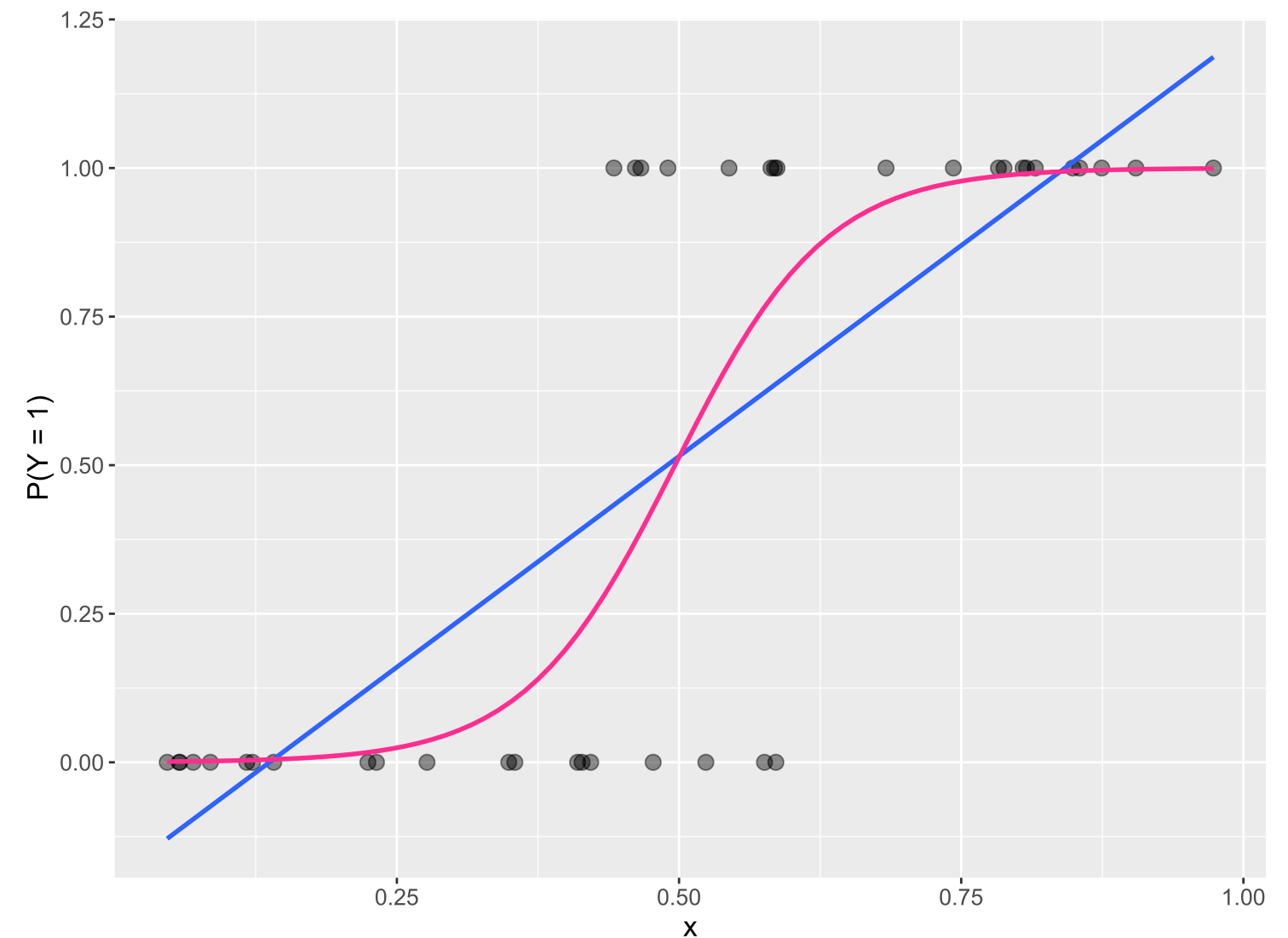
$Y \sim \text{Bernoulli}(p)$ where $p = P(Y = 1) = P(\text{success})$.

Explanatory variables: Can be a mix of categorical and quantitative

Goal: Build a model for $P(Y = 1)$.

Why can't we use linear regression?

- Regression line = estimated **probability** of success
- For valid values of x , we are predicting the probability is less than 0 or greater than 1.
- The estimated probabilities based on the logistic regression model are bounded between 0 and 1.



What does the **logistic regression model** look like?

Logistic Regression Model

$$\log \left(\frac{P(Y = 1)}{1 - P(Y = 1)} \right) = \beta_o + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$$

Left hand side has many interpretations:

$$\begin{aligned} \text{LHS} &= \log \left(\frac{P(Y = 1)}{1 - P(Y = 1)} \right) \\ &= \log (\text{odds (of success)}) \\ &= \text{logit}(P(Y = 1)) \end{aligned}$$

Note:

$$\text{odds} = \frac{\text{prob of success}}{\text{prob of failure}}$$

Example

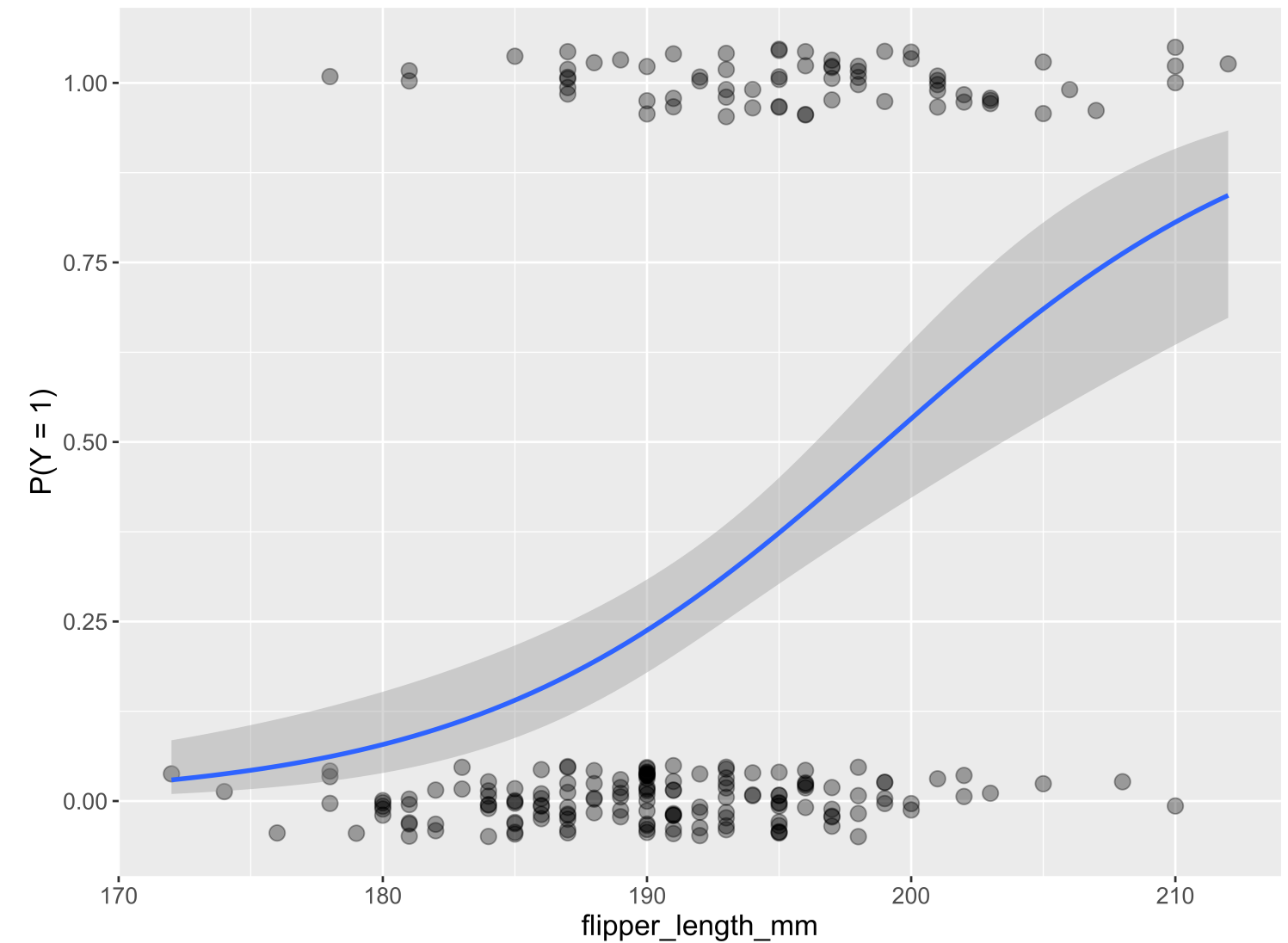
Let's return to the Palmer Penguins dataset and build a model that predicts whether a penguin is a **Chinstrap** or a **Adelie**.

```
1 library(palmerpenguins)
2
3 penguins2 <- penguins %>%
4   filter(species != "Gentoo") %>%
5   mutate(species_num = if_else(species == "Chinstrap",
6                                1, 0)) %>%
7   drop_na(species_num, flipper_length_mm)
8
9 count(penguins2, species)
```

```
# A tibble: 2 × 2
  species      n
  <fct>    <int>
1 Adelie    151
2 Chinstrap   68
```

Example

```
1 ggplot(data = penguins2,  
2       mapping = aes(x = flipper_length_mm,  
3                     y = species_num)) +  
4   geom_jitter(size = 3, alpha = 0.4,  
5             width = 0, height = 0.05) +  
6   labs(y = "P(Y = 1)") +  
7   geom_smooth(method = 'glm',  
8             method.args =  
9             list(family = "binomial"))
```



Example

```
1 mod <- glm(species ~ flipper_length_mm, family = "binomial",  
2           data = penguins2)  
3 library(broom)  
4 tidy(mod)
```

A tibble: 2 × 5

	term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>
1	(Intercept)	-25.8	4.90	-5.26	0.000000141
2	flipper_length_mm	0.129	0.0253	5.12	0.000000299

But what do the coefficient estimates $\hat{\beta}_0 = -25.8$ and $\hat{\beta}_1 = 0.129$ represent?

Interpreting Coefficients

$$\exp(\hat{\beta}_1) = \frac{\text{odds of being Chinstrap for flipper length } t + 1 \text{ mm}}{\text{odds of being Chinstrap for flipper length } t \text{ mm}}$$

```
1 exp(0.129)
```

```
[1] 1.13769
```

The odds that a penguin in Chinstrap instead of Adelie is 1.138 times the odds of a penguin whose flipper is 1 mm shorter.

Inference

Is flipper length a useful explanatory variable?

```
1 tidy(mod)
```

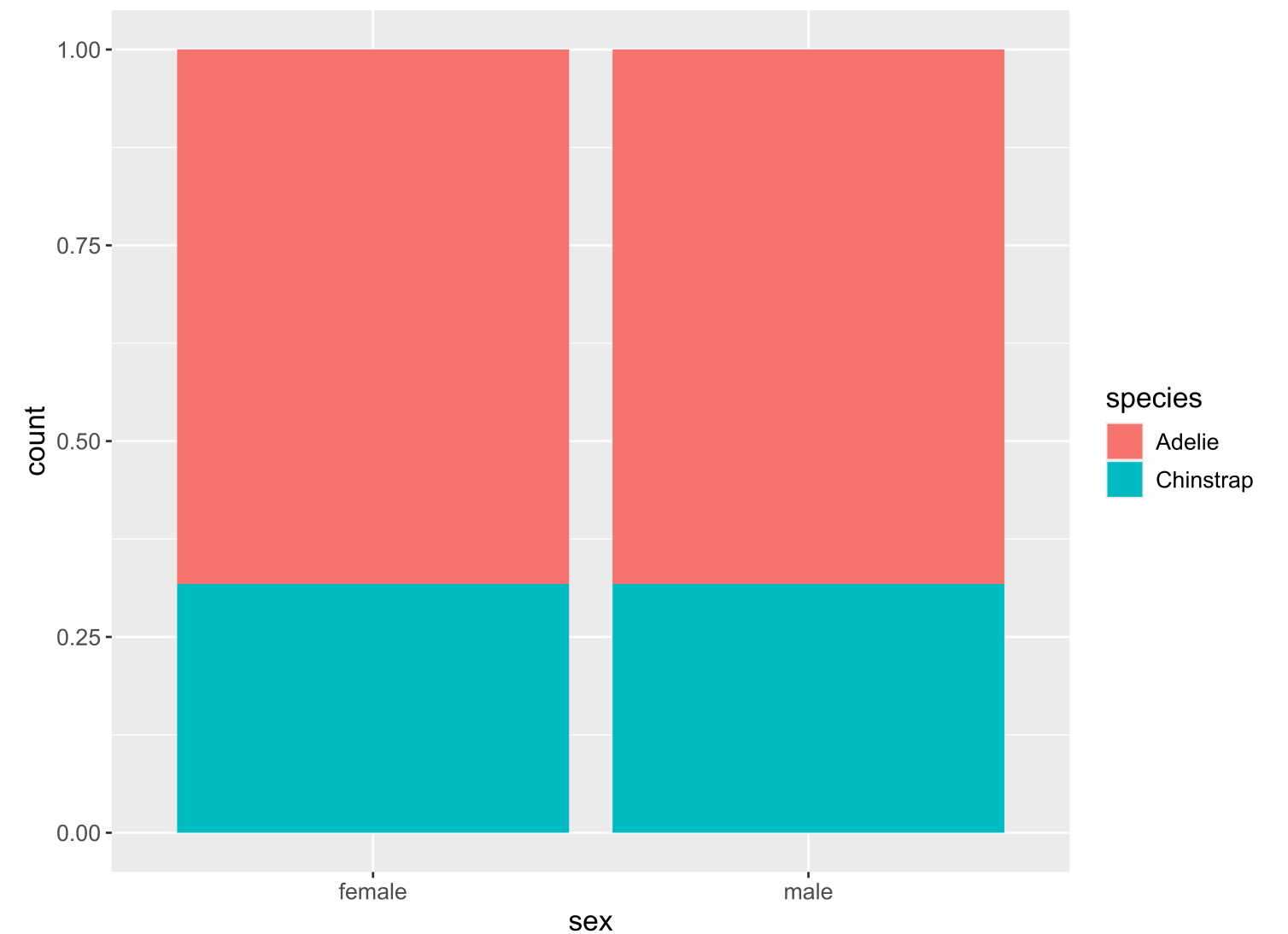
A tibble: 2 × 5

	term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>
1	(Intercept)	-25.8	4.90	-5.26	0.000000141
2	flipper_length_mm	0.129	0.0253	5.12	0.000000299

Is a Penguin's Sex a Good Predictor of Species?

Can also try a categorical predictor.

```
1 penguins2 %>%  
2   drop_na(sex) %>%  
3   ggplot(mapping = aes(x = sex,  
4                         fill = species)) +  
5   geom_bar(position = "fill")
```



Logistic Regression with a Categorical Explanatory Variable

```
1 mod2 <- glm(species ~ sex, family = "binomial",  
2             data = penguins2)  
3 tidy(mod2)
```

A tibble: 2 × 5

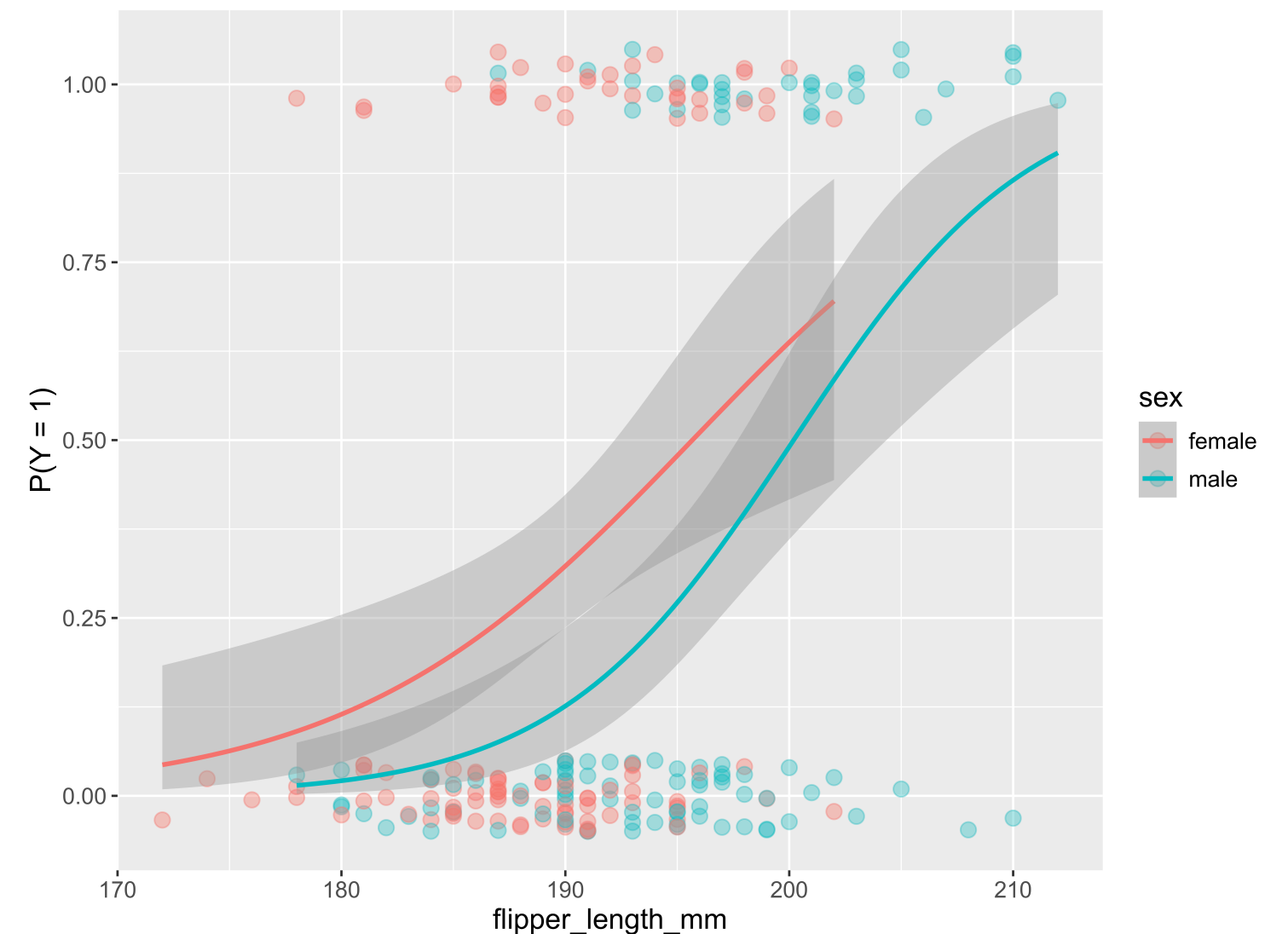
	term	estimate	std.error	statistic	p.value
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	(Intercept)	-7.64e- 1	0.208	-3.68e+ 0	0.000233
2	sexmale	2.51e-15	0.294	8.56e-15	1.000

Is sex a useful explanatory variable?

Multiple Logistic Regression

Similar to linear regression, can have a mix of categorical and quantitative predictors.

```
1 penguins2 %>%
2   drop_na(sex) %>%
3   ggplot(mapping = aes(x = flipper_length_mm,
4                         y = species_num,
5                         color = sex)) +
6   geom_jitter(size = 3, alpha = 0.4,
7               width = 0, height = 0.05) +
8   labs(y = "P(Y = 1)") +
9   geom_smooth(method = 'glm',
10              method.args =
11                list(family = "binomial"))
```



Multiple Logistic Regression

```
1 mod3 <- glm(species ~ flipper_length_mm + sex, family = "binomial",  
2             data = penguins2)  
3 tidy(mod3)
```

A tibble: 3 × 5

	term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>
1	(Intercept)	-31.5	5.73	-5.49	0.0000000391
2	flipper_length_mm	0.162	0.0300	5.39	0.0000000722
3	sexmale	-0.982	0.373	-2.64	0.00840

```
1 1/exp(-0.982)
```

```
[1] 2.66979
```

- Controlling for flipper length, the odds that a female penguin is a Chinstrap instead of an Adelie is 2.67 times the odds a male penguin is a Chinstrap instead of an Adelie.

New Data Example

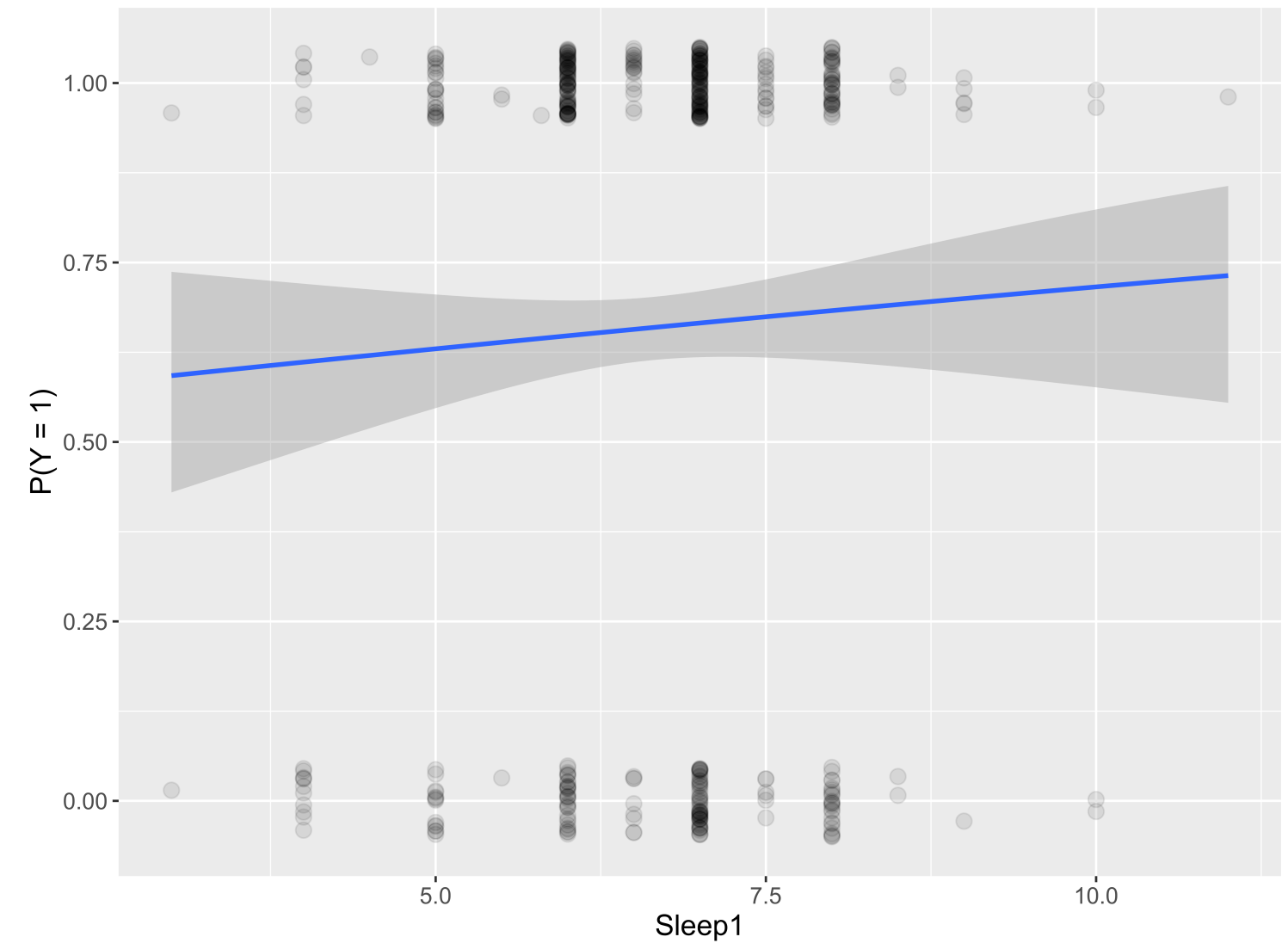
Let's look at the **PA Seniors** dataset, which is a sample of PA high school seniors in 2019. Can we build a model to predict whether or not a student will select “happy” as the answer to the question “When you grow up, would you prefer to be famous, happy, healthy, or rich?”

```
1 library(Lock5Data)
2 data("PASeniors2019")
3
4 PASeniors <- PASeniors2019 %>%
5   drop_na(WorkHours, Preference) %>%
6   filter(Preference != "") %>%
7   mutate(PreferenceH = case_when(
8     Preference == "Happy" ~ "Picked Happy",
9     Preference != "Happy" ~ "Something Else"
10  ),
11  PreferenceH_num = case_when(
12    Preference == "Happy" ~ 1,
13    Preference != "Happy" ~ 0
14  ))
15
16 count(PASeniors, PreferenceH)
```

	PreferenceH	n
1	Picked Happy	295
2	Something Else	152

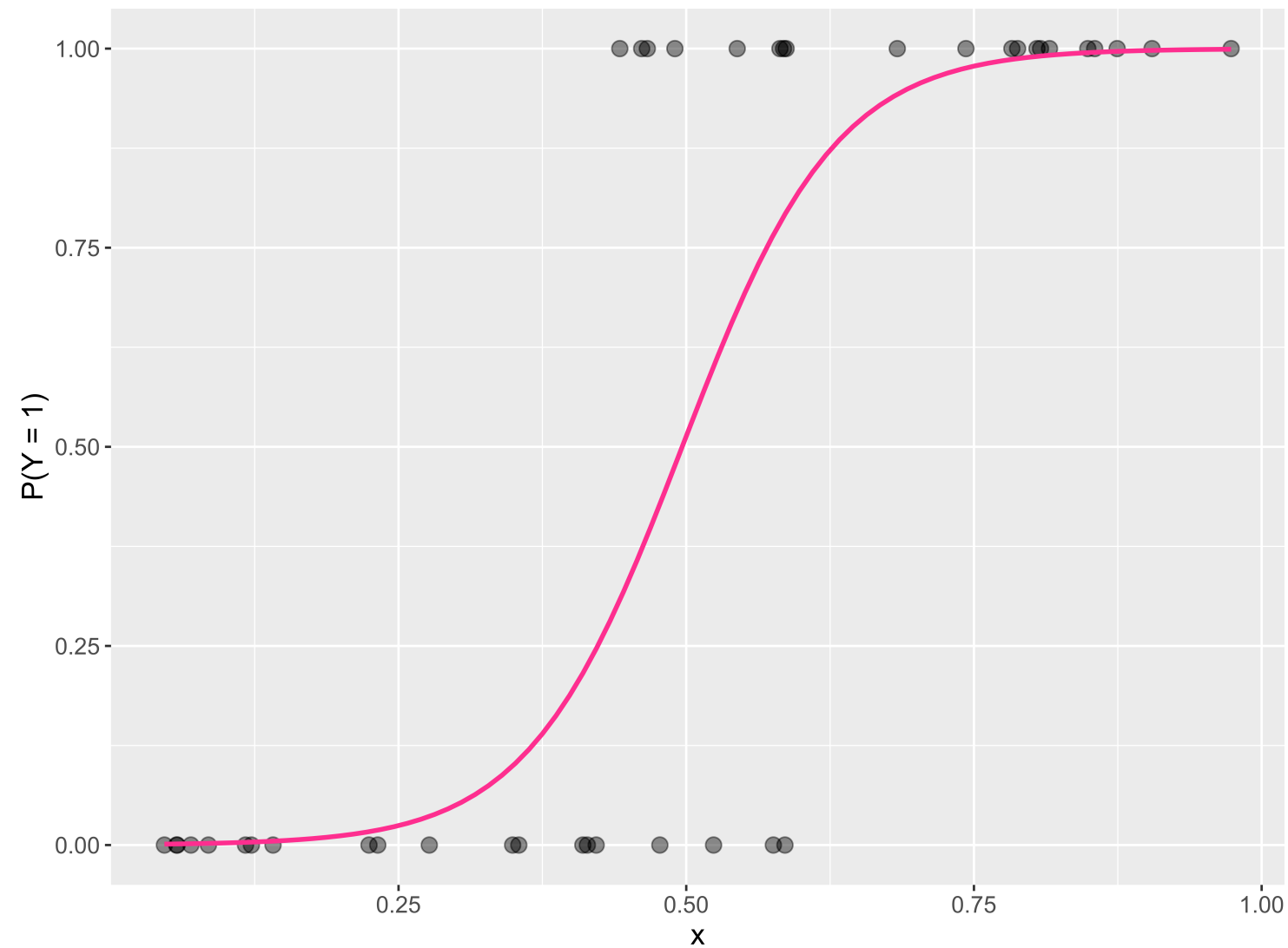
Example

```
1 ggplot(data = PASeniors,  
2       mapping = aes(x = Sleep1,  
3                     y = PreferenceH_num)) +  
4   geom_jitter(size = 3, alpha = 0.1,  
5             width = 0, height = 0.05) +  
6   labs(y = "P(Y = 1)") +  
7   geom_smooth(method = 'glm',  
8             method.args =  
9             list(family = "binomial"))
```

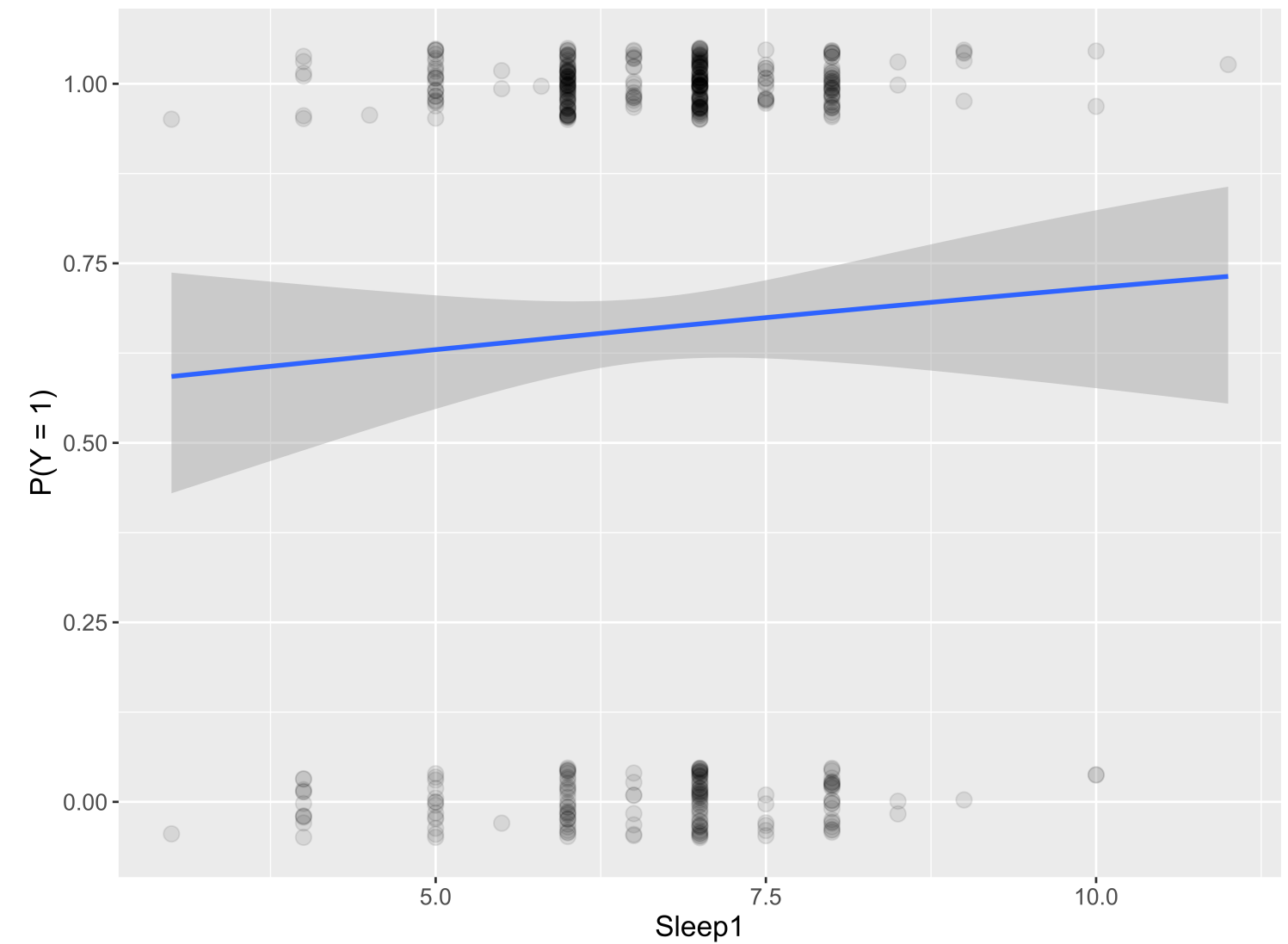


Example

This is what an helpful explanatory variable looks like!



This is what an unhelpful explanatory variable looks like!



Logistic Regression Table

```
1 happy_mod <- glm(PreferenceH_num ~ Sleep1, data = PASeniors)
2 library(broom)
3 tidy(happy_mod)
```

```
# A tibble: 2 × 5
```

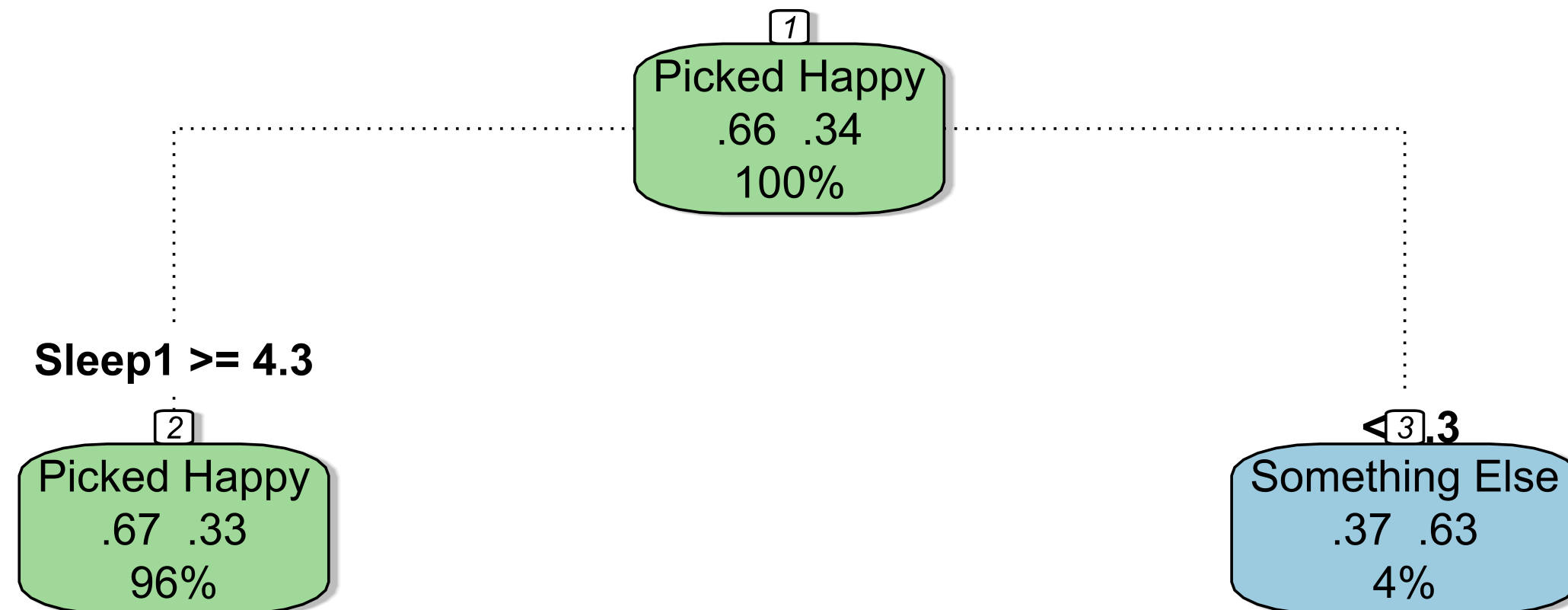
	term	estimate	std.error	statistic	p.value
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	(Intercept)	0.542	0.134	4.04	0.0000626
2	Sleep1	0.0177	0.0199	0.890	0.374

I tried several variables and struggled to build a useful logistic regression model.

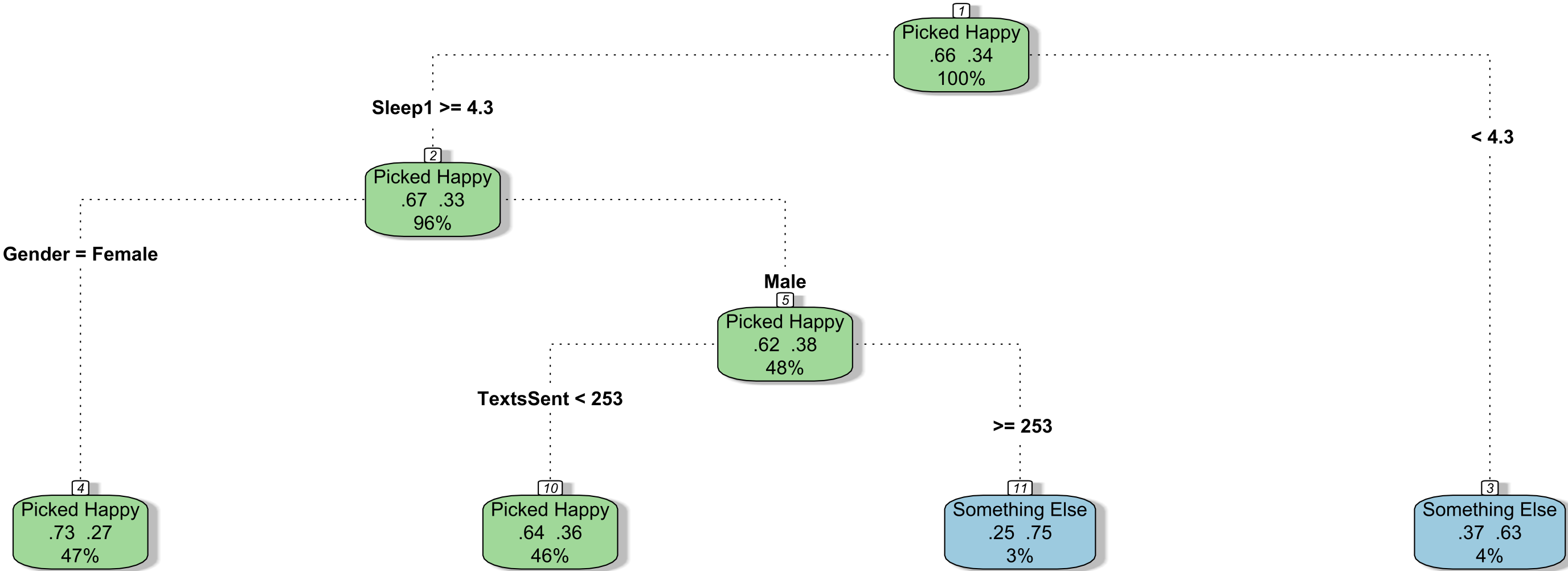
There are LOTS of models out there beyond regression! Let's return to tree models.

Classification Trees

- Split and split and split... sample into two groups based on an explanatory variable.
 - Wants groups to be homogeneous as possible for the response variable.
- **Stop** splitting when it is no longer very predictively useful to do so.

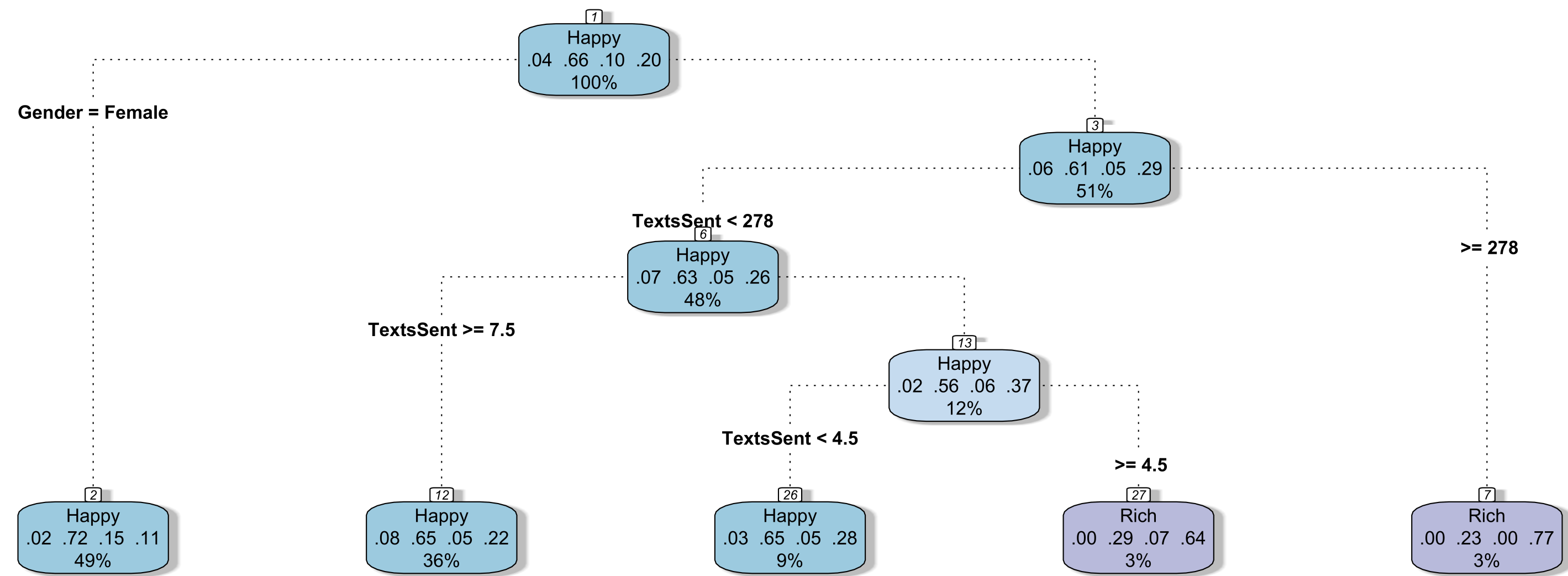


Classification Trees



Classification Trees

Can handle more than two categories for the response variable!



Coding the Classification Tree

```
1 # Fit the tree
2 pref_tree <- rpart(Preference ~ Sleep1 + GetToSchool + TextsSent + VideoGameHours + Gender,
3                   data = PASeniors, method = "class",
4                   control = rpart.control(cp = 0, minsplit = 2, minbucket = 5))
5
6 # Look at the performance at different cp values
7 printcp(pref_tree)
8
9 # Prune the tree based on a specific cp
10 pref_tree_pruned <- prune(pref_tree, cp = 0.01315789 + 0.00001)
11
12 # Graph the tree
13 fancyRpartPlot(pref_tree_pruned, cex = .8, caption = "", type = 4)
```

Installing R and RStudio